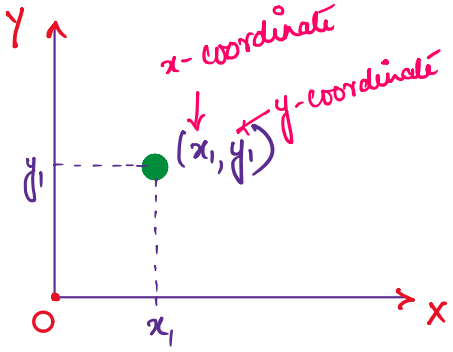
**Mathematical Foundations**

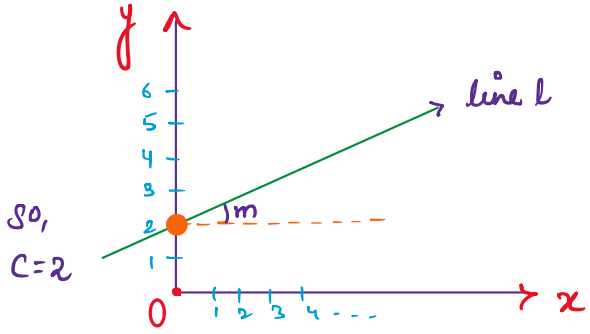
## ★ Linear Algebra :-

### Equation of a line :-

$$y = mx + c$$

where, $y$ = y-coordinate
$x$ = x-coordinate
$m$ = slope of the line
$c$ = y-intercept



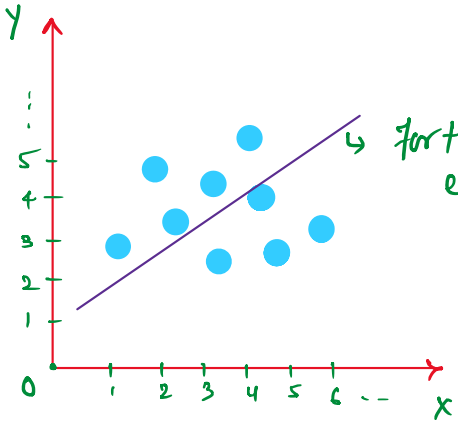




m = ∠ b/w line l & horizontal axis
c = point at which the line intersects y-axis

## Why Linear Algebra?

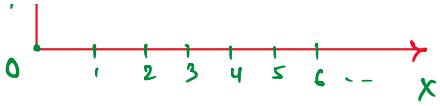
In regression problem →

| exp. | Salary |
|---|---|
| $x_1$ | $y_1$ |
| $x_2$ | $y_2$ |
| ⋮ | ⋮ |



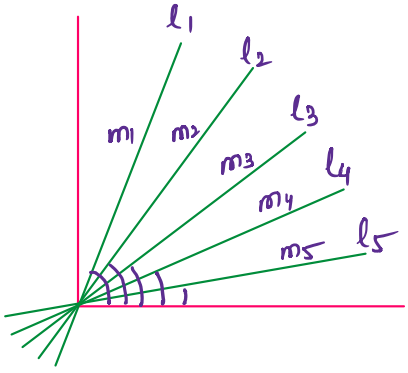

↳ for this we study eq. of a line

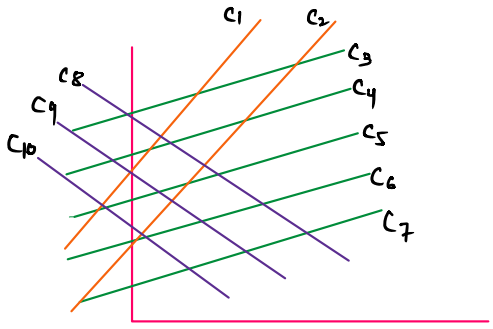





* Can a line be drawn like the above one only?

Ans. No, there can be multiple lines having different angles & intercepts



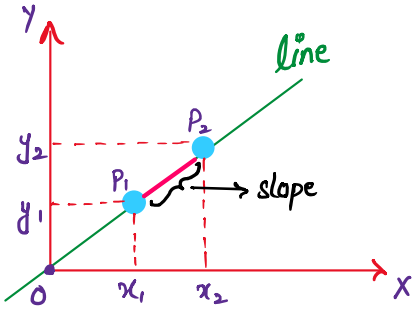

As m & c can have any value, therefore these values along with the value of x decides y. Hence, y is a dependent variable.

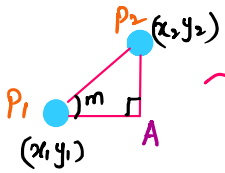
Ques. How to calculate the value of m (slope)?

Ans.

Slope = Ratio of vertical change to the horizontal change b/w two points.

$$\text{Slope} = \frac{\text{Vertical Change}}{\text{Horizontal Change}}$$




Acc. to Trigonometry ⟵ 

⟶ Right ∠ Triangle

To calculate the angle b/w slope & horizontal line, we use →

$$\tan\theta = \frac{P}{B}$$

relating this formula with

$$\tan\theta = \frac{P}{B}$$ relating this formula with the figure

or $$\tan\theta = \frac{P_2 A}{P_1 A}$$

$$\tan\theta = \frac{y_2 - y_1}{x_2 - x_1}$$

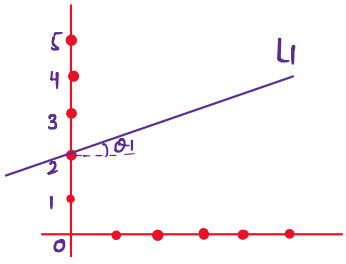
or $\tan\theta = m$

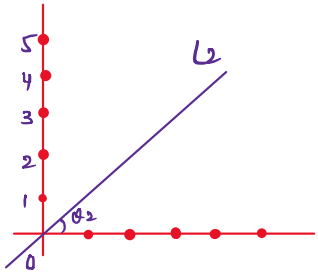
So,

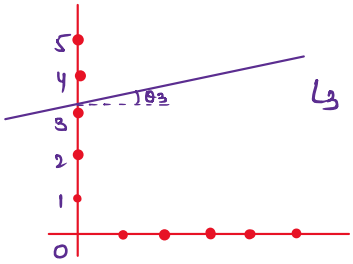
$$\boxed{m = \frac{y_2 - y_1}{x_2 - x_1}}$$

∴ This is how the value of m is calculated.

★ Combining slope & intercept :-




Here, $m_1 = \tan\theta_1$, $c_1 = 2$ | $m_2 = \tan\theta_2$, $c_2 = 0$ | $m_3 = \tan\theta_3$, $c_3 = 3.1$

Relating this to ML

Algorithms like → Linear Regression ⇝ Best fitting line

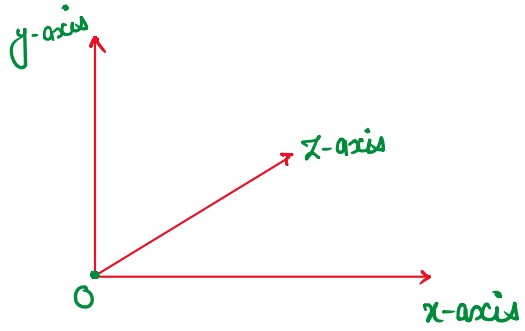
↳ Logistic Regression ⇝ Best separating line

So, whatever we have studied above is specific to 2 dimensions

So, whatever we have studied above is specific to 2 dimensions
(x-axis, y-axis)

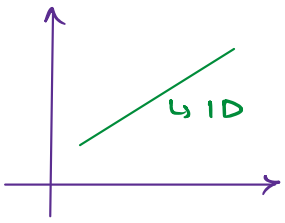
Ques. But what about higher dimensions ?

3D :-

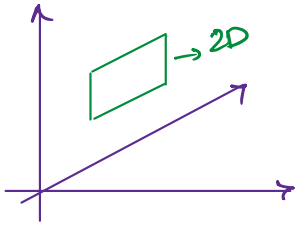



eq. in case of 3D :-

$$y = m_1 x + m_2 z + c$$




2D graph




3D graph

In general way,

$$y = m_1 x_1 + m_2 x_2 + c$$ ⇝ eq. of a plane

& for nD :-

$$y = m_1 x_1 + m_2 x_2 + m_3 x_3 + m_4 x_4 + \text{-----} + c$$

↳ eq. of a hyperplane

## Relating dimensions with the data :-

Ex-

Data →

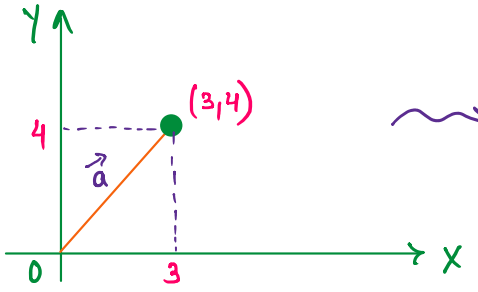
| → features | | | | |
|---|---|---|---|---|
| F1 | F2 | F3 | F4 | F5 |
| | | | | |

No. of features = No. of dimensions

## ★ Vectors :-

- Object having both magnitude & direction.
- In mathematics, vectors are representation of points
- It is represented by $\vec{v}$

In coordinate geometry,




$\vec{a} = [3, 4]$

$\vec{a_x}$ , $\vec{a_y}$ [2 components]

∴ $\vec{a} = \vec{a_x}, \vec{a_y}$

where, $\vec{a_x}$ = horizontal component

$\vec{a_y}$ = vertical "

3D representation :-

$\vec{a} = \vec{a_x}, \vec{a_y}, \vec{a_z}$

nD representation :-

$$\vec{a} = \vec{a_x}, \vec{a_y}, \vec{a_z} \text{ ---- }, \vec{a_n}$$

* Vector form of a line or plane or hyperplane :-

Line → $mx+c$

$[m]$ $[x]$

Plane → $m_1x_1 + m_2x_2 + c$

$\begin{bmatrix} m_1 \\ m_2 \end{bmatrix}$ $\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$ ⇝ This is a vector in Maths

Hyperplane → $m_1x_1 + m_2x_2 + m_3x_3 + \text{ -- --- } + m_nx_n$

$$M = \begin{bmatrix} m_1 \\ m_2 \\ m_3 \\ \vdots \\ m_n \end{bmatrix}, \quad X = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_n \end{bmatrix}$$

The above eq. will become →

$$\boxed{y = MX + c}$$

↳ Generalized Equation